

Math 63: Real Analysis

Prishita Dharampal

Credit Statement: Talked to Sair Shaikh'26, and Math Stack Exchange.

Problem 1. Discuss the continuity of the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} 0 & \text{if } x \text{ is irrational,} \\ \frac{1}{q} & \text{if } x = \frac{p}{q} \text{ with } p, q \in \mathbb{Z} \text{ and } \gcd(p, q) = 1. \end{cases}$$

Solution.

The function f is not continuous at rational points, but is continuous at irrational points.

1. At irrational points.

Let x be an irrational number. Take any sequence (x_n) such that,

$$\lim_{n \rightarrow \infty} x_n = x$$

Since x is irrational $f(x) = 0$. For each n the term x_n is either rational or irrational:

- (a) x_n is irrational.

Then $f(x_n) = 0$.

- (b) x_n is rational.

Let $x_n = \frac{p_n}{q_n}$ in lowest terms. Then, we claim that as $x_n \rightarrow x$, $q_n \rightarrow \infty$.

Assume for the sake of contradiction that q_n does not tend to infinity, as x_n tends to x . That is suppose, $1 \leq q_n \leq N$ for some N . Then, by definition of limits, for every ϵ , there exist some M such that $n > M$, $d(x_n, x) < \epsilon$. That is,

$$x_n = \frac{p_n}{q_n} \in (x - \epsilon, x + \epsilon)$$

Which is,

$$\begin{aligned} x - \epsilon &< \frac{p_n}{q_n} < x + \epsilon \\ q_n(x - \epsilon) &< p_n < q_n(x + \epsilon) \end{aligned}$$

Since the interval is bounded, there are only a finite number of integer values p_n can have, and we already know that there are only a finite number of denominators that x_n since q_n is bounded by N by assumption. Since there are finitely many possible rational numbers in the interval $(x - \epsilon, x + \epsilon)$ and infinitely many terms of the sequence lie in that interval, by the pigeonhole principle, at least one rational number must occur infinitely often. Hence a subsequence converges to that rational number. But limits are unique, and by assumption the limit x is irrational. Hence a contradiction! That is, $q_n \rightarrow \infty$ as $x_n \rightarrow x$.

Note that as $x_n \rightarrow x$,

$$q_n \rightarrow \infty \implies f(x_n) = \frac{1}{q_n} \rightarrow 0$$

Either ways, $f(x_n) \rightarrow 0 = f(x)$, therefore f is continuous at every irrational point.

2. At rational points.

Let x be a rational number, where $x = \frac{p}{q}$ in lowest terms, then $f(x) = \frac{1}{q}$. Since the irrationals are dense in the reals, there exist a sequence (x_n) of irrational x_n such that

$$\lim_{n \rightarrow \infty} x_n = x$$

But for all x_n , $f(x_n) = 0$. That is,

$$\lim_{n \rightarrow \infty} f(x_n) = 0$$

But because $f(x) = \frac{1}{q} \neq 0$,

$$\lim_{n \rightarrow \infty} f(x_n) \neq f(\lim_{n \rightarrow \infty} x_n)$$

Hence, the function is not continuous at rational points (by Proposition on Page 74).

Problem 2. Let E, E' be metric spaces, $f : E \rightarrow E'$ a continuous function. Prove that if E is compact and f is one-one onto then $f^{-1} : E' \rightarrow E$ is continuous. (Hint: f sends closed sets onto closed sets, therefore open sets onto open sets.)

Solution.

Assume $f : E \rightarrow E'$ is a bijective continuous function, and E is compact.

Note that since f is bijective, f^{-1} exists and sends $f(p)$ to p for all $p \in E$. To show that f^{-1} is continuous we need to show that for any open set U in E , $(f^{-1})^{-1}(U) \subset E'$ is open. That is it suffices to show that f sends open sets to open sets.

Let $C \subset E$ be a closed subset. All closed subsets of a compact metric space are compact, hence C is compact. And because continuous functions send compact sets to compact sets, $f(C) \subset E'$ is also compact. All compact subsets of a metric space are closed (Corollary 3, Page 56), hence $f(C)$ is also closed. That is f maps closed sets to closed sets.

Note that since f is bijective and maps closed sets C to closed sets $f(C)$, it would map their complements (open sets) C' map to the complements of their images $f(C)'$ (also open sets). Hence, f sends open sets to open sets, and f^{-1} is continuous.

Problem 3. A metric space E is said to be arcwise connected if, given any $p, q \in E$, there is a continuous function $f : [0, 1] \rightarrow E$ such that $f(0) = p, f(1) = q$. Show that any connected open subset of E^n is arcwise connected.

Solution.

Let C be a connected open subset of E^n . Pick $p \in C$. Let $C_p \subset C$ be defined as follows,

$$C_p := \{q \in C \mid \exists f : [0, 1] \rightarrow E^n : f(0) = p, f(1) = q\}$$

Note that C_p is non-empty since $p \in C_p$. Now, consider some $q \in C_p$. Since C is open, there exists some ϵ such that $B_\epsilon(q) \subset C$ is open. For any point $q' \in B_\epsilon(q)$, there exists a line segment between $q = (q_1, \dots, q_n)$ and $q' = (q'_1, \dots, q'_n)$ to be the set of points

$$\{(q_1) + (q'_1 - q_1)t, \dots, q_n + (q'_n - q_n)t : t \in [0, 1]\} \subset B_\epsilon(q)$$

By joining these segments (p to q , and q to q') we get a continuous segment from p to q' . That is, $B_\epsilon(q) \subset C_p$. Since q was an arbitrary point in C_p this holds for all points in C_p , and C_p is open.

Now consider a point q such that $q \in C \setminus C_p$. Then, since C is open, there exists some ϵ such that $B_\epsilon(q)$ is open. Then, $B_\epsilon(q) \cap C_p = \emptyset$. (For the sake of contradiction assume the intersection isn't empty, then $\exists x \in B_\epsilon(q) \cap C_p$ such that there is a path from p to x and there is also a straight line from x to q , that is $q \in C_p$ which is a contradiction.) Since q was an arbitrary point in $C \setminus C_p$, we can see that $C \setminus C_p$ is also open. Given C is connected, C_p is non-empty and both C_p and its complement $C \setminus C_p$ are open, C_p must be the whole subset C .

Hence, any connected open subset $C \subset E^n$ is arcwise connected.

Problem 4. Let $f, f_1, f_2, f_3, \dots$ be continuous real-valued functions on the compact metric space E , with $\lim_{n \rightarrow \infty} f_n = f$. Prove that if $f_1(p) \leq f_2(p) \leq f_3(p) \leq \dots$ for all $p \in E$ then the sequence $f_1, f_2, f_3, \dots$ converges uniformly.

Solution.

We know that the sequence (f_n) converges to f . That is, for any p and for any $\epsilon > 0$, there exists a $N_p > 0$ such that $d(f(p), f_n(p)) = |f(p) - f_n(p)| < \epsilon$, whenever $n > N_p$.

We want to show that given any $\epsilon > 0$, there is a positive integer N such that $d(f(p), f_n(p)) = |f(p) - f_n(p)| < \epsilon$ whenever $n > N$, for all $p \in E$.

To find the supremum of the N_p s we define functions $g_n : E \rightarrow \mathbb{R}$, such that

$$g_n(p) = f(p) - f_n(p)$$

Since continuous real valued functions on a nonempty compact metric space attain a maximum at some point, let $M_n = \sup\{g_n(p), \forall p \in E\}$. Note that the sequence (M_n) converges to 0 because the sequence of functions (f_n) is monotonic increasing. That is,

$$\sup\{g_n(p), \forall p \in E\} \geq \sup\{g_{n+1}(p), \forall p \in E\}$$

So (M_n) is monotonic decreasing and bounded by zero. Note that $\lim_{n \rightarrow \infty} M_n = 0$ since, by definition of the limit of (f_n) we can get arbitrarily close to f .

Then, for any $\epsilon > 0$, there exists an N , such that $M_n < \epsilon$, for all $n > N$, and

$$d(f(p), f_n(p)) \leq M_n < \epsilon$$

for all $p \in E$. Hence, (f_n) is uniformly continuous.

Problem 5. If E is a compact metric space and $p_0 \in E$ we get a real-valued function F on $C(E)$ by setting $F(f) = f(p_0)$ for all $f \in C(E)$. Prove that F is uniformly continuous.

Solution.

The space $C(E)$ is space of all real valued continuous functions, with the metric

$$d = \max\{|f(p) - g(p)| \mid \forall p \in E\}$$

Note that we have

$$\begin{aligned} F : C(E) &\rightarrow \mathbb{R} \\ f &\mapsto f(p_0) \end{aligned}$$

To show that F is uniformly continuous we need to show that $\forall \epsilon > 0$ there exists a $\delta > 0$ such that if $f, g \in C(E)$ and $d(f, g) < \delta$ then $|f(p_0) - g(p_0)| < \epsilon$.

Pick $\delta = \epsilon$. Now, if $d(f, g) < \delta$, then we have

$$d(f, g) = \max\{|f(p) - g(p)| \mid \forall p \in E\} < \delta = \epsilon$$

Note that

$$|f(p_0) - g(p_0)| \leq \max\{|f(p) - g(p)| \mid \forall p \in E\} < \delta = \epsilon$$

That is,

$$|f(p_0) - g(p_0)| < \epsilon$$

Hence, F is uniformly continuous.